

COMPONENT-1

(Operating Systems with
Linux as Case Study)

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Operating Systems with Linux as Case Study (Component I)

Question :

Compare Round Robin, Shortest Job First (Pre-emptive) and Priority Scheduling (Pre-emptive) algorithms with one example having a minimum of 12 processes, in terms of Average Turn Around time, Waiting Time and Response Time.

Introduction

CPU scheduling is the process of selecting a process from the ready queue and allocating the CPU to it. The objective is to use the CPU efficiently while minimizing waiting and response times.

Here, we compares

- Round Robin
- Shortest Job First (Pre-emptive)
- Priority Scheduling (Pre-emptive)

The algorithms are compared using Average TurnAround Time, Waiting Time and Average Response Time.

Important Terms

- Arrival Time (AT): Time at which a process enters the ready queue.
- Burst Time (BT): CPU time required by a process
- Completion Time (CT): Time at which a process finishes
- Turnaround Time (TAT): $TAT = CT - AT$
- Waiting Time (WT): $WT = TAT - BT$
- Response Time (RT): $RT = \text{First CPU start time} - AT$

The Process set

The following 12 processes are used for all the three algorithms.

Process	AT	BT	Priority
P1	0	8	2
P2	1	4	1
P3	2	9	3
P4	3	5	2
P5	4	2	4
P6	5	7	1

P7	6	3	5
P8	7	6	3
P9	8	4	2
P10	9	5	4
P11	10	2	1
P12	11	3	3

Assumption : For Priority scheduling , a smaller priority number means higher priority.

For Round Robin , the Time Quantum = 3 units

Round Robin Scheduling

Round Robin is a pre-emptive scheduling algorithm in which every process gets a fixed amount of CPU time called the time quantum. If the process does not finish within the quantum, it is moved to the end of the ready queue.

Time Quantum = 3 units

Gantt Chart

P1	P2	P3	P4	P1	P5	P6	P7	P2	P8	P9	P10	P3	P11	P12	P4	P1	P6	P8	P9	P10	P3	P6	
0	3	6	9	12	15	17	20	23	24	27	30	33	36	38	41	43	45	48	51	52	54	57	58

Process	CT	TAT	WT	RT
P1	45	45	37	0
P2	24	23	19	2
P3	57	55	46	4
P4	43	40	35	6
P5	17	13	11	11
P6	58	53	46	12
P7	23	17	14	14
P8	51	44	38	17
P9	52	41	40	19
P10	54	45	40	21
P11	38	28	26	26
P12	41	30	27	27

Average Turn Around Time (TAT) = 36.42 units

Average Waiting Time (WT) = 31.58 units

Average Response Time (RT) = 13.25 units

Shortest Job First (Pre-emptive)

Pre-emptive SJF is also called Shortest Remaining Time First (SRTF). The process having the shortest remaining CPU burst time is selected. If a new process arrives with a shorter remaining time, the currently running process is pre-empted.

Gantt Chart

P1	P2	P5	P7	P11	P12	P9	P4	P10	P8	P1	P6	P3	
0	1	5	7	10	12	15	19	24	29	35	42	49	58

Process	First Start	CT	TAT	WT	RT
P1	0	42	42	34	0
P2	1	5	4	0	0
P3	49	58	56	47	47
P4	19	24	21	16	16
P5	5	7	3	1	1
P6	42	49	44	37	37
P7	7	10	4	1	1
P8	29	35	28	22	22

P9	15	19	11	7	7
P10	24	29	20	15	15
P11	10	12	2	0	0
P12	12	15	4	1	1

Average TurnAround Time = 19.92 units

Average Waiting Time = 15.08 units

Average Response Time = 12.25 units

Priority Scheduling (Pre-emptive)

In pre-emptive Priority scheduling, every process is assigned a priority. The CPU is allocated to the process having the highest priority. A newly arrived higher-priority process can pre-empt the currently running process.

Here, smaller priority number = higher priority.

Grant Chart

P1	P2	P6	P11	P1	P4	P9	P3	P8	P12	P5	P10	P7	
0	1	5	12	14	21	26	30	39	45	48	50	55	58

Process	Firststart	CT	TAT	WT	RT
P1	0	21	21	13	0
P2	1	5	4	0	0
P3	30	39	37	28	28
P4	21	26	23	18	18
P5	48	50	46	44	44
P6	5	12	7	0	0
P7	55	58	52	49	49
P8	39	45	38	32	32
P9	26	30	22	18	18
P10	50	55	46	41	41
P11	12	14	4	2	2
P12	45	48	37	34	34

Average TurnAround Time = 28.08 units

Average Waiting Time = 23.25 units

Average Response Time = 22.17 units

Final Comparison

- SJF (Pre-emptive) gives the lowest Average TurnAround Time.
- SJF (Pre-emptive) also gives the lowest Avg. Waiting Time.

- SJF (Pre-emptive) gives the lowest Average Response Time for this particular process set.
- Round Robin provides comparatively fair CPU allocation and is suitable for time-sharing systems but its turnaround and waiting times are higher in this example.
- Priority Scheduling is useful when some processes must be given preference, but lower-priority processes can experience starvation.
- SJF can also cause starvation for long processes if short processes keep arriving.

Algorithms	Avg. TAT	Avg. WT	Avg. RT
Round Robin	36.42	31.58	13.25
SJF (Pre-emptive)	19.92	15.08	12.25
Priority Scheduling (Pre-emptive)	28.08	23.25	22.17

Advantages and Disadvantages

Algorithms	Advantages	Disadvantages
Round Robin	Fair, good response for interactive systems, prevents starvation.	Performance depends heavily on time quantum.
SJF (Pre-emptive)	Low waiting and turnaround time	Long processes may starve, burst time must be estimated.
Priority (Pre-emptive)	Important processes gets CPU first.	Low priority processes may starve

Conclusion

From the above activity, Shortest Job First (SJF) Pre-emptive performs best for the given set of processes because it produces the lowest average turnaround time, waiting time and response time. However, there is ~~no~~ no single algorithm that is best for every situation. Round Robin is more suitable

For interactive and time-sharing systems, while Priority scheduling is useful when processes have different levels of importance.

Best for this example: SJF (Pre-emptive)

Best for fairness: Round Robin

Best when process

importance matters: Priority scheduling (Pre-emptive)